

WEEK 6 OVERVIEW

An aerial photograph of a residential neighborhood that has been completely inundated with floodwater. The water is a murky, brownish-grey color, covering the streets, yards, and surrounding areas. Numerous houses with various roof colors (brown, grey, red) are visible, partially submerged. Green trees are scattered throughout the landscape, their lower branches and leaves underwater. A central street runs vertically through the middle of the frame, with a few vehicles visible in the distance. The overall scene depicts a severe flooding event.

James Doss-Gollin

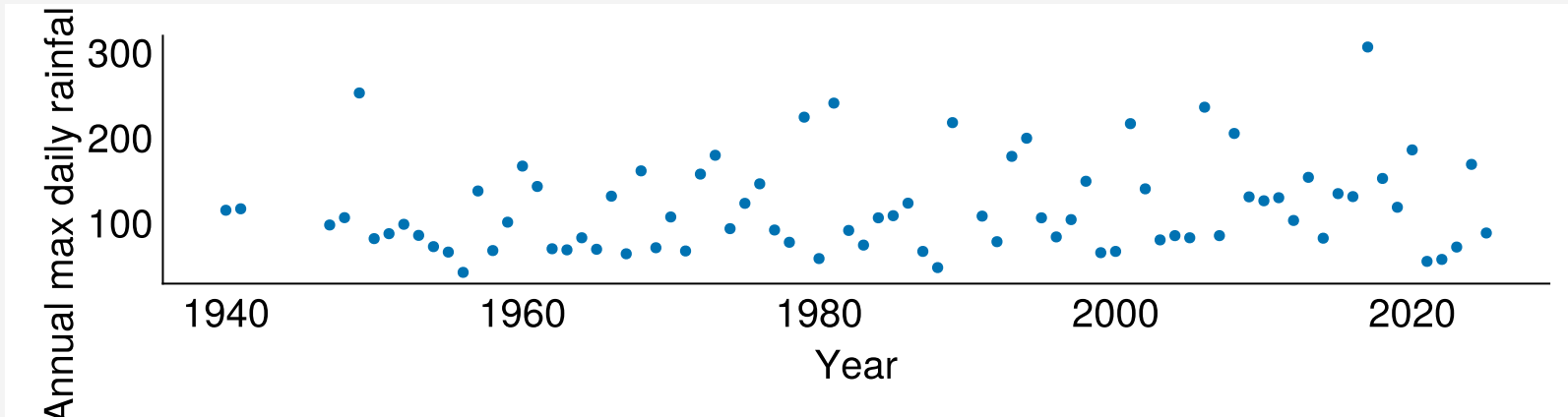
CEVE 543

2026-09-28

THIS WEEK

- **Monday:** nonstationarity, trend tests, and covariates
- **Wednesday:** Smith et al. (2026); bring your two questions
- **Friday:** lab 6, fitting nonstationary extreme value models

ANNUAL MAXIMUM RAINFALL AT HOUSTON HOBBY

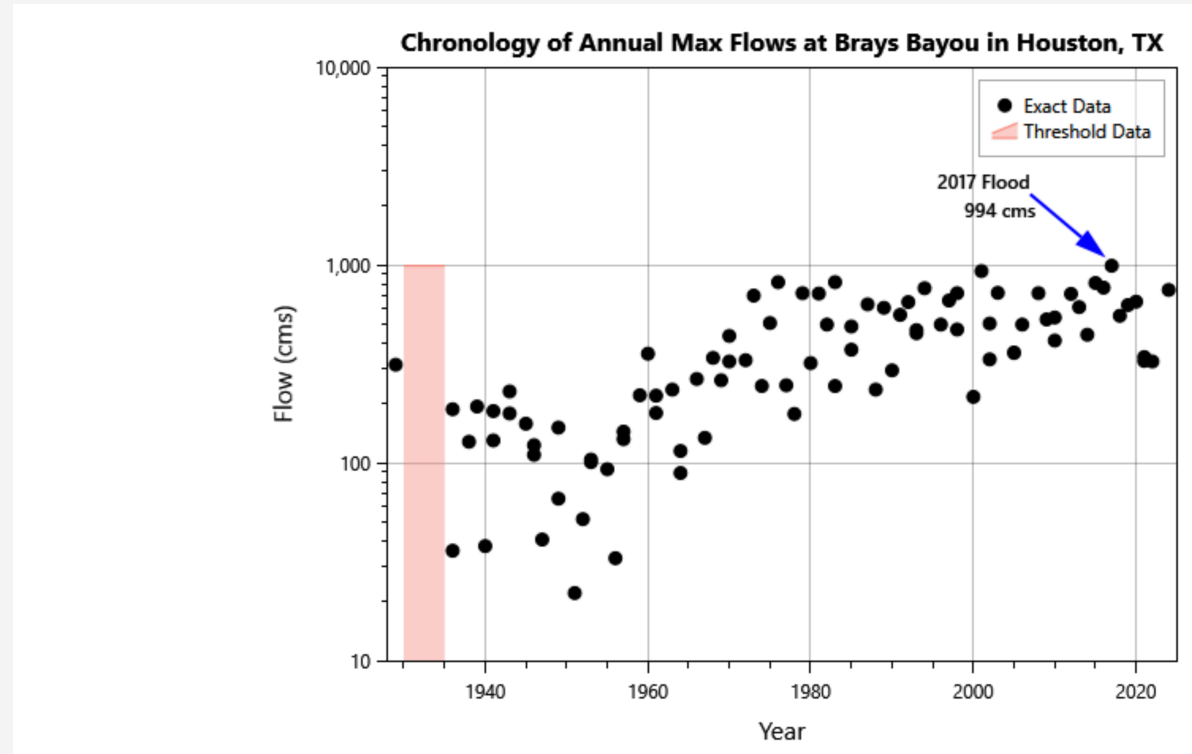


Houston Hobby Airport (GHCN-Daily USW00012918), 80 years with at least 350 days of data.

💡 Think

Last week you subtracted mean sea level from the Galveston record before fitting. What would you subtract here?

A TREND MEANS THE YEARS ARE NOT IDENTICALLY DISTRIBUTED



Brays Bayou at Houston, annual peak flow, 1929-2024 (Smith et al., 2026, Fig. 2, CC BY 4.0).

SUBTRACTING A TREND IS NOT ENOUGH

- Galveston: sea level adds to surge, so subtracting it shifts the distribution and keeps its shape.
- Brays Bayou: after about 1968 the typical flood rose and the spread, on a log scale, narrowed (Smith et al., 2026, pp. 14, 24).
- No single number subtracted from each year removes both changes.

STATIONARY GIVEN A COVARIATE

Let each year's distribution depend on a covariate x_t , such as sea level or CO2:

$$Y_t \mid x_t \sim \text{GEV}(\mu(x_t), \sigma(x_t), \xi)$$

- μ , σ , and ξ are the location, scale, and shape from week 4.
- Given the covariate, the years are independent: **conditional independence**.
- Subtracting mean sea level at Galveston was the special case $\mu(x_t) = \mu_0 + x_t$, with σ and ξ fixed.
- Whether a record is “stationary” depends on what you condition on.